# BigRAG Cinema – Tentative Benchmark Results

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## Overview

Benchmark results from 5 Amazon Reviews 2023 categories totaling approximately 20 million records. Four hybrid query execution strategies benchmarked at 5 data fractions (10%–100%) using Apache Spark in local mode on a 16-CPU / 64GB SLURM node.

| Dataset | Total Rows | Queries | Repetitions |
|---|---|---|---|
| All Beauty | 693,548 | 50 | 3 |
| Amazon Fashion | 2,473,302 | 50 | 2 |
| Appliances | 2,103,927 | 50 | 2 |
| Arts & Crafts | 8,859,715 | 30 | 1 |
| Baby Products | 5,962,394 | 30 | 1 |
| **Total** | **20,092,886** | | |

Table 1: Benchmark dataset summary from the Amazon Reviews 2023 corpus.

| Strategy | All Beauty (694K) | Amazon Fashion (2.5M) | Appliances (2.1M) | Arts & Crafts (8.9M) | Baby Products (6.0M) |
|---|---|---|---|---|---|
| *Median Latency (ms)* | | | | | |
| Filter-First | **1,245** | 4,652 | 3,728 | 17,126 | 13,047 |
| Vector-First | **13,435** | 48,002 | 41,046 | 172,191 | 119,110 |
| Hybrid Parallel | **13,440** | 47,923 | 40,827 | 171,459 | 115,428 |
| Adaptive | **1,289** | 4,508 | 3,735 | 16,619 | 12,489 |
| *P99 Latency (ms)* | | | | | |
| Filter-First | **9,352** | 33,574 | 27,069 | 111,673 | 70,001 |
| Vector-First | **13,556** | 48,210 | 41,323 | 172,827 | 121,202 |
| Hybrid Parallel | **13,580** | 48,194 | 41,062 | 172,547 | 116,679 |
| Adaptive | **13,553** | 48,342 | 41,127 | 171,667 | 118,556 |
| *Mean Throughput (qps)* | | | | | |
| Filter-First | **0.69** | 0.19 | 0.23 | 0.05 | 0.07 |
| Vector-First | **0.07** | 0.02 | 0.02 | 0.01 | 0.01 |
| Hybrid Parallel | **0.07** | 0.02 | 0.02 | 0.01 | 0.01 |
| Adaptive | **0.66** | 0.19 | 0.22 | 0.05 | 0.07 |

Table 2: Cross-dataset strategy comparison at full data scale. Bold indicates best strategy per dataset per metric.

| Strategy | All Beauty | | Amazon Fashion | | Appliances | | Arts & Crafts | | Baby Products | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 10% (ms) | Factor | 10% (ms) | Factor | 10% (ms) | Factor | 10% (ms) | Factor | 10% (ms) | Factor |
| Filter-First | 173 | 7.2× | 484 | 9.6× | 423 | 8.8× | 1,595 | 10.7× | 1,193 | 10.9× |
| Vector-First | 1,380 | 9.7× | 4,820 | 10.0× | 4,133 | 9.9× | 17,031 | 10.1× | 11,601 | 10.3× |
| Hybrid Parallel | 1,407 | 9.5× | 4,839 | 9.9× | 4,149 | 9.8× | 17,134 | 10.0× | 11,595 | 10.0× |
| Adaptive | 199 | 6.5× | 508 | 8.9× | 448 | 8.3× | 1,645 | 10.1× | 1,172 | 10.7× |

Table 3: Scalability analysis: median latency at 10% data fraction and growth factor to full scale (100%). Lower factor = better scalability.

| Comparison | All Beauty | Amazon Fashion | Appliances | Arts & Crafts | Baby Products |
|---|---|---|---|---|---|
| vs. Vector-First | 10.8× | 10.3× | 11.0× | 10.1× | 9.1× |
| vs. Hybrid Parallel | 10.8× | 10.3× | 11.0× | 10.0× | 8.8× |
| vs. Adaptive | 1.0× | 1.0× | 1.0× | 1.0× | 1.0× |

Table 4: Speedup of Filter-First over other strategies (median latency ratio at full scale). Higher = Filter-First is faster by that factor.

| Metric | All Beauty | Amazon Fashion | Appliances | Arts & Crafts | Baby Products |
|---|---|---|---|---|---|
| Median Latency | Filter-First | Adaptive | Filter-First | Adaptive | Adaptive |
| P99 Latency | Filter-First | Filter-First | Filter-First | Filter-First | Filter-First |
| Mean Throughput | Filter-First | Adaptive | Filter-First | Adaptive | Adaptive |
| Latency Variance | Vector-First | Vector-First | Hybrid Parallel | Vector-First | Hybrid Parallel |
| **Total Wins** | Filter-First (3/4) | Adaptive (2/4) | Filter-First (3/4) | Adaptive (2/4) | Adaptive (2/4) |

Table 5: Best-performing strategy per dataset at full scale. Latency Variance measured as P99/P50 ratio (lower = more consistent).

| Strategy | Fraction | P50 | P95 | P99 | Min | Max |
|---|---|---|---|---|---|---|
| Filter-First | 10% | 1,595 | 8,215 | 11,160 | 1,574 | 11,844 |
| | 25% | 3,924 | 20,200 | 27,079 | 3,844 | 28,642 |
| | 50% | 8,086 | 41,292 | 55,473 | 7,616 | 58,660 |
| | 75% | 12,740 | 62,364 | 83,605 | 11,446 | 88,445 |
| | 100% | 17,126 | 82,351 | 111,673 | 15,430 | 118,225 |
| Vector-First | 10% | 17,031 | 17,115 | 17,126 | 16,954 | 17,129 |
| | 25% | 42,602 | 42,783 | 42,919 | 42,419 | 42,964 |
| | 50% | 85,572 | 85,860 | 85,908 | 85,329 | 85,925 |
| | 75% | 128,581 | 128,956 | 129,215 | 128,127 | 129,296 |
| | 100% | 172,191 | 172,741 | 172,827 | 171,458 | 172,861 |
| Hybrid Parallel | 10% | 17,134 | 17,245 | 17,290 | 17,014 | 17,306 |
| | 25% | 42,625 | 42,925 | 42,946 | 42,477 | 42,950 |
| | 50% | 85,458 | 85,823 | 85,907 | 84,877 | 85,932 |
| | 75% | 127,855 | 128,236 | 128,306 | 127,251 | 128,325 |
| | 100% | 171,459 | 172,195 | 172,547 | 170,553 | 172,680 |
| Adaptive | 10% | 1,645 | 17,332 | 17,414 | 1,598 | 17,425 |
| | 25% | 4,029 | 43,270 | 43,504 | 3,863 | 43,546 |
| | 50% | 8,341 | 85,409 | 85,607 | 7,680 | 85,660 |
| | 75% | 12,819 | 128,097 | 128,235 | 11,521 | 128,285 |
| | 100% | 16,619 | 171,282 | 171,667 | 15,254 | 171,730 |

Table 6: Latency percentile breakdown (ms) – Arts & Crafts.

| Strategy | Fraction | Median Lat. (ms) | P95 Lat. (ms) | Throughput (qps) | Results |
|---|---|---|---|---|---|
| Filter-First | 10% | 1,595 | 8,215 | 0.54 | 3.3 |
| | 25% | 3,924 | 20,200 | 0.22 | 3.3 |
| | 50% | 8,086 | 41,292 | 0.11 | 3.3 |
| | 75% | 12,740 | 62,364 | 0.07 | 3.3 |
| | 100% | 17,126 | 82,351 | 0.05 | 3.3 |
| Vector-First | 10% | 17,031 | 17,115 | 0.06 | 1.9 |
| | 25% | 42,602 | 42,783 | 0.02 | 1.9 |
| | 50% | 85,572 | 85,860 | 0.01 | 1.8 |
| | 75% | 128,581 | 128,956 | 0.01 | 1.8 |
| | 100% | 172,191 | 172,741 | 0.01 | 1.7 |
| Hybrid Parallel | 10% | 17,134 | 17,245 | 0.06 | 3.3 |
| | 25% | 42,625 | 42,925 | 0.02 | 3.3 |
| | 50% | 85,458 | 85,823 | 0.01 | 3.3 |
| | 75% | 127,855 | 128,236 | 0.01 | 3.3 |
| | 100% | 171,459 | 172,195 | 0.01 | 3.3 |
| Adaptive | 10% | 1,645 | 17,332 | 0.51 | 3.3 |
| | 25% | 4,029 | 43,270 | 0.21 | 3.3 |
| | 50% | 8,341 | 85,409 | 0.10 | 3.3 |
| | 75% | 12,819 | 128,097 | 0.07 | 3.3 |
| | 100% | 16,619 | 171,282 | 0.05 | 3.3 |

Table 7: Detailed benchmark results – Arts & Crafts.

| Strategy | All Beauty | Digital Music | Mean |
|---|---|---|---|
| Filter-First | **1.0000** | **1.0000** | **1.0000** |
| Vector-First | 0.8100 | 0.9700 | 0.8900 |
| Hybrid Parallel | **1.0000** | **1.0000** | **1.0000** |
| Adaptive | **1.0000** | **1.0000** | **1.0000** |
| *By Selectivity Bucket (first dataset)* | | | |
| Filter-First | Low: 1.0 | Med: 1.0 | High: 1.0 |
| Vector-First | Low: 1.0 | Med: 1.0 | High: 0.3667 |
| Hybrid Parallel | Low: 1.0 | Med: 1.0 | High: 1.0 |
| Adaptive | Low: 1.0 | Med: 1.0 | High: 1.0 |

Table 8: Recall@10 per strategy. Filter-First serves as ground truth (recall=1.0). Bold indicates perfect recall.

## Figures

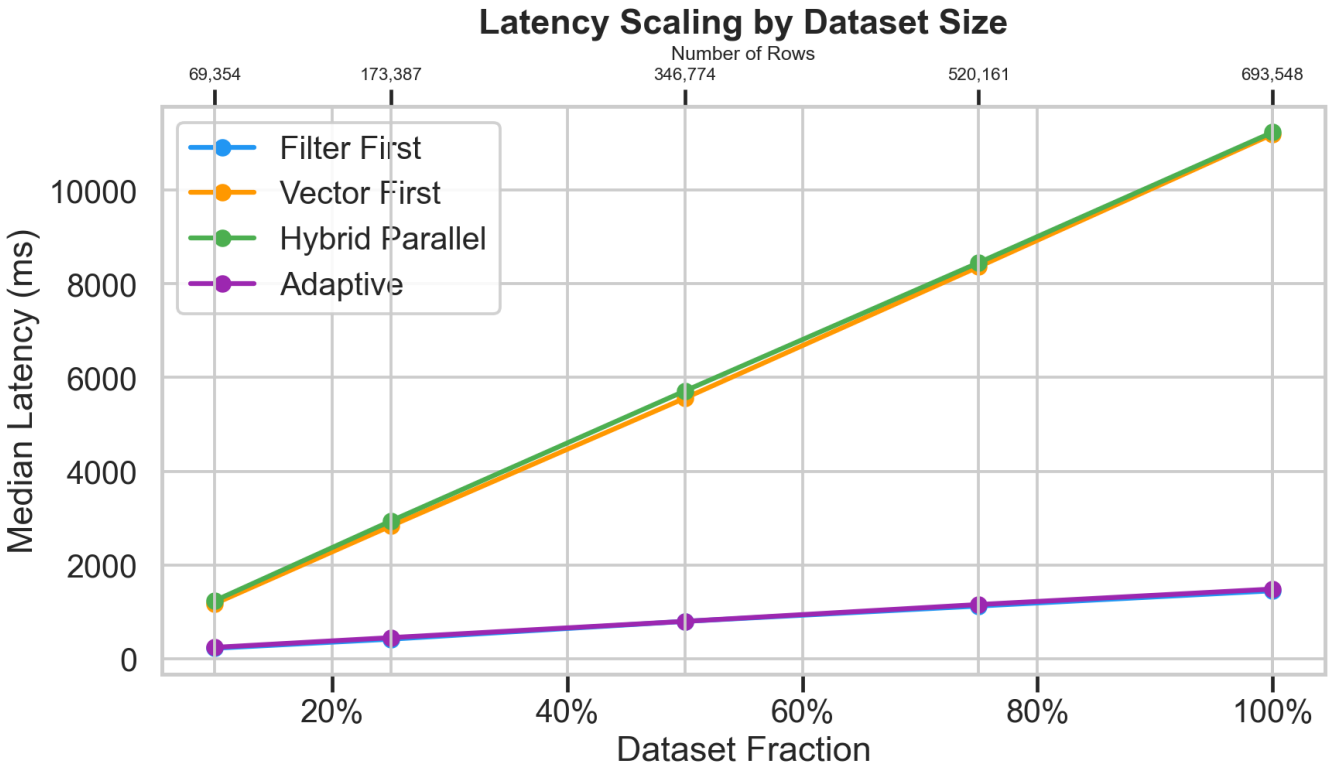


Figure 1: Latency scaling curves across data fractions.

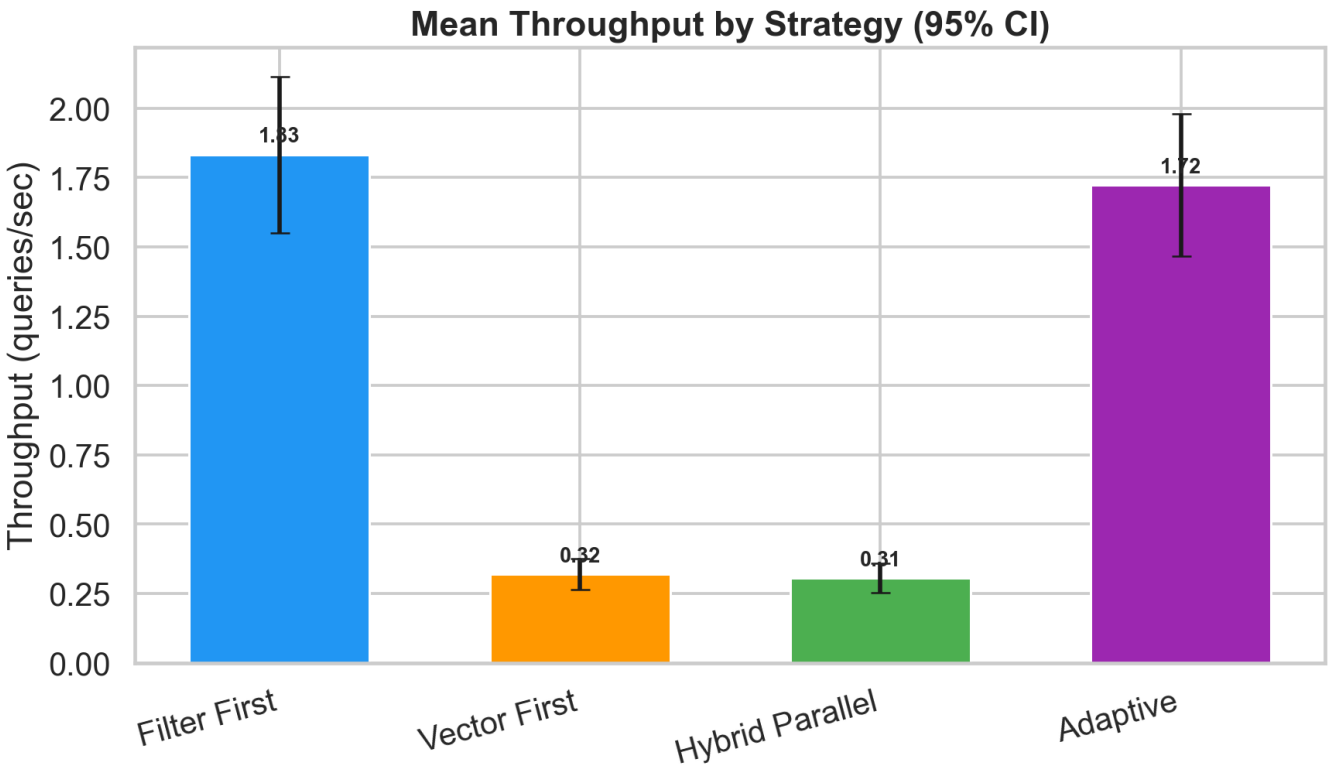


Figure 2: Throughput comparison by strategy.

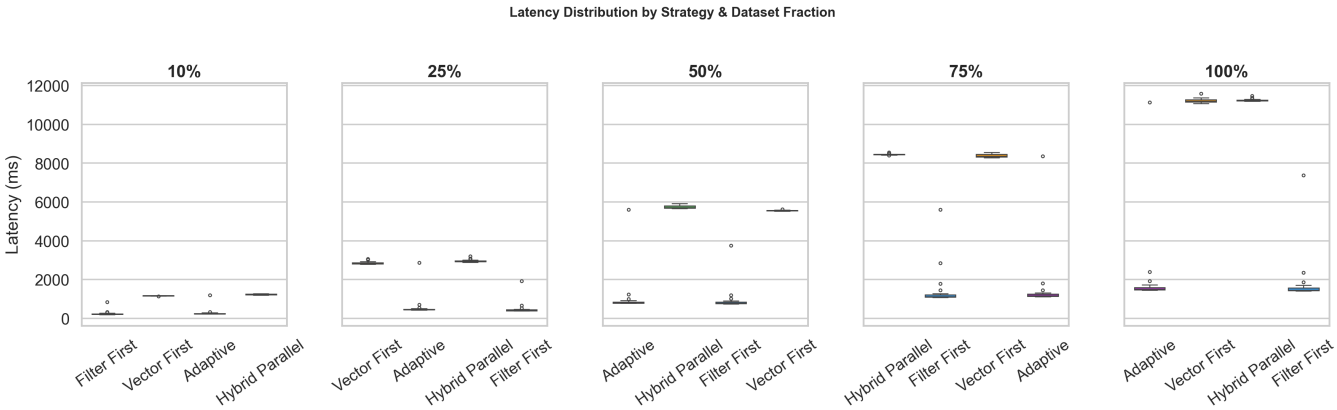


Figure 3: Latency distribution boxplots.

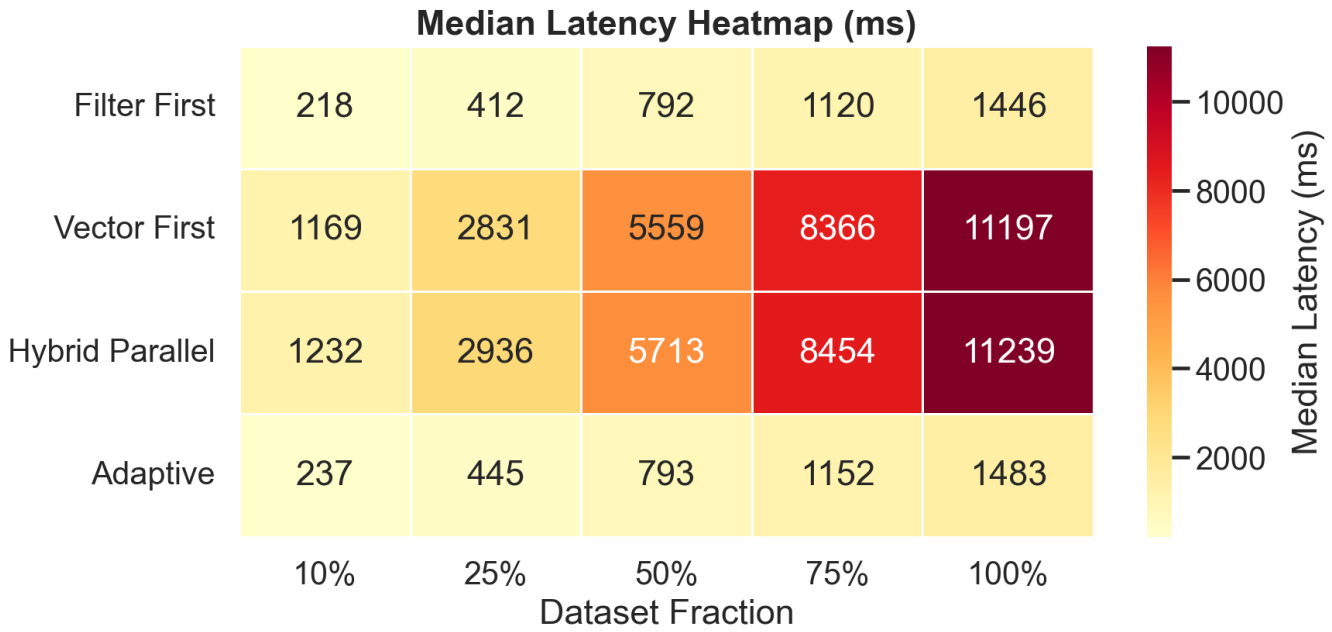


Figure 4: Latency heatmap across strategies and fractions.

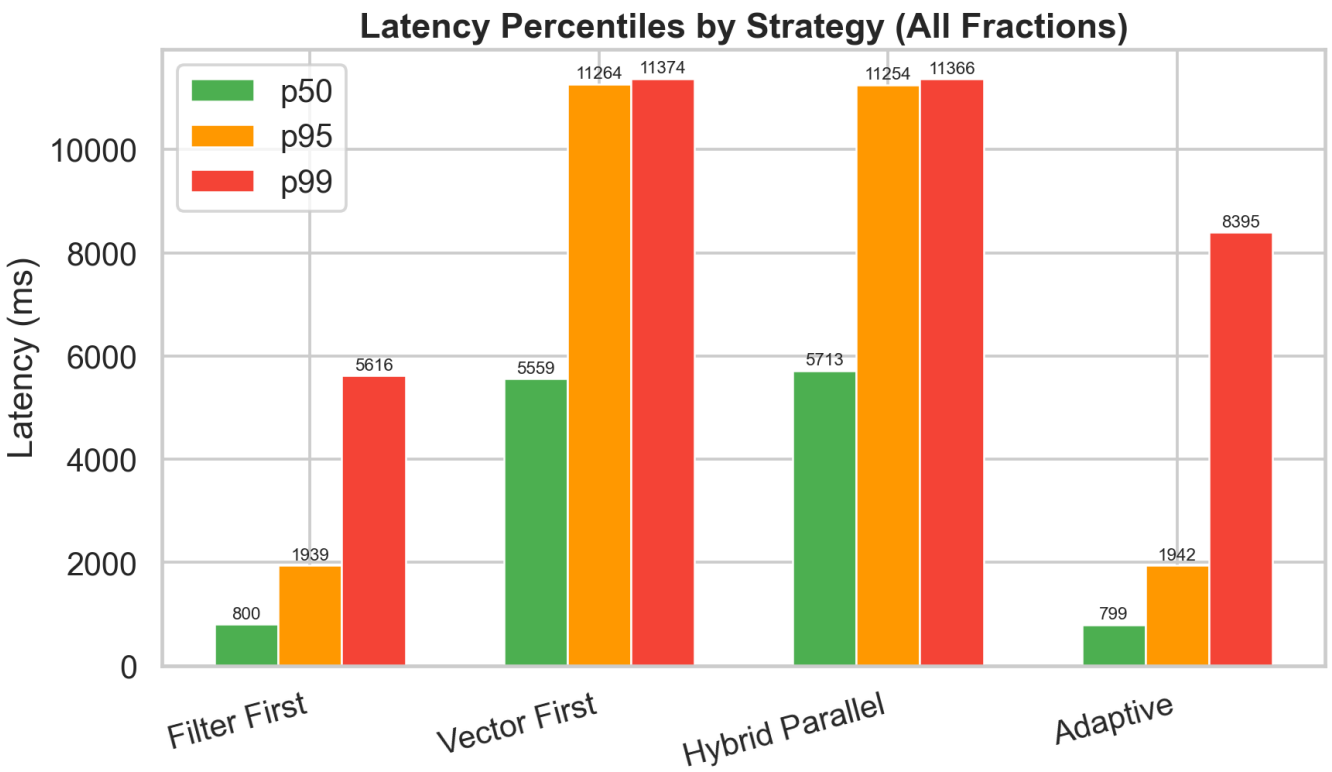


Figure 5: Latency percentile bars (P50, P95, P99).

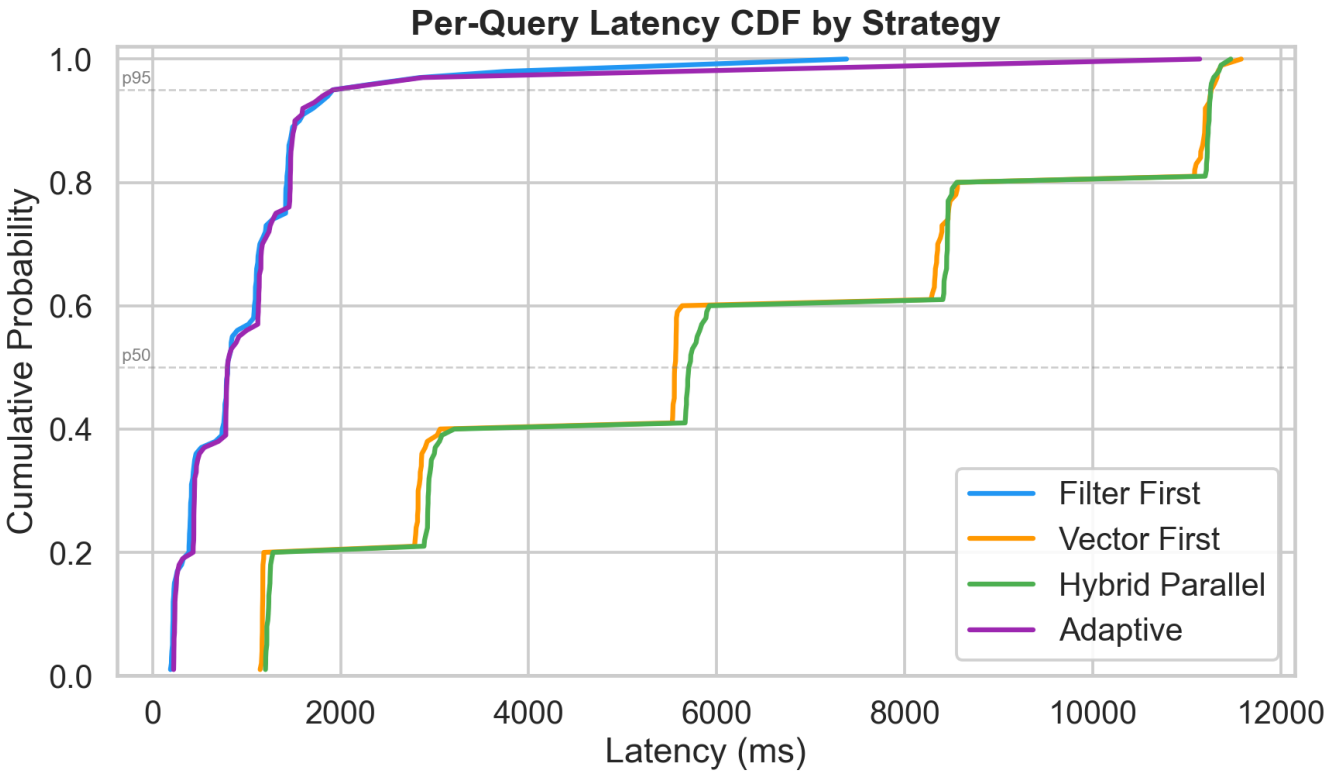


Figure 6: Latency CDF across strategies.

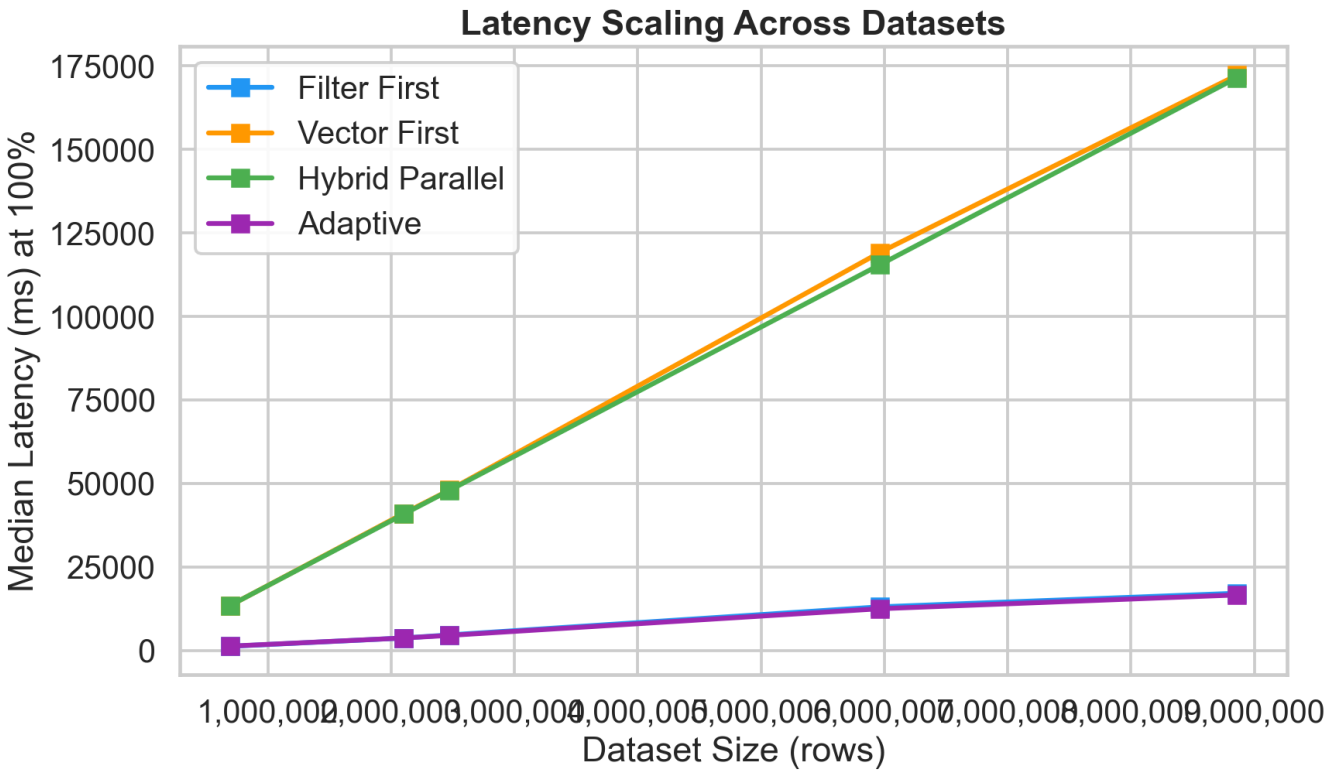


Figure 7: Cross-dataset latency scaling comparison.

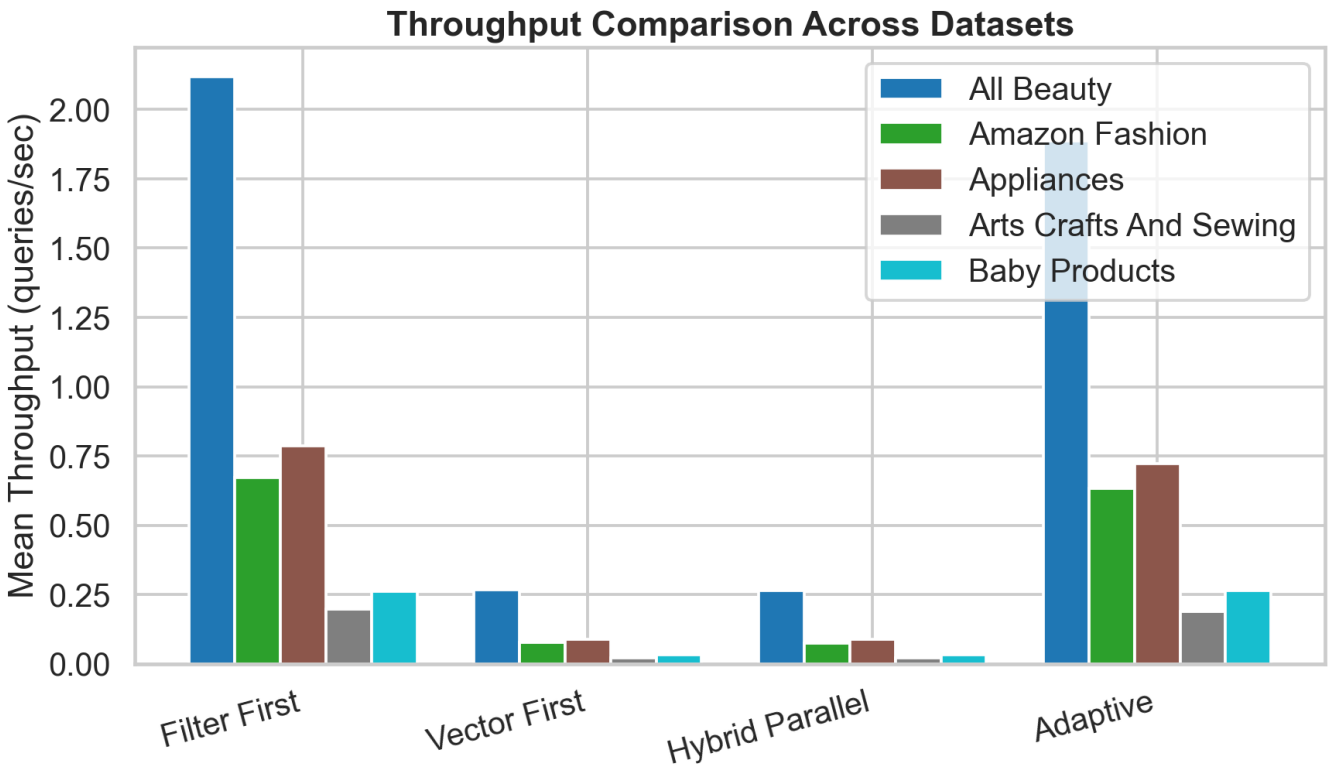


Figure 8: Cross-dataset throughput comparison.